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ORIGINAL RESEARCH

Molecular docking, linear and nonlinear QSAR studies on factor Xa inhibitors

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Abstract

Factor Xa (FXa) enzyme has an important role in the blood coagulation system. Disruption in the enzyme function results in the production of blood clots. Therefore, inhibition of the factor Xa with anticoagulant drugs is an important target in thromboembolic therapy. Experimental design of new drugs is a time-consuming and expensive process. Application of molecular modeling is an essential step for designing new and high yield drugs to achieve better results. In this study, 36 aroylguanidine derivatives that inhibit destructive activities by binding to the FXa enzyme were selected. Quantitative Structure-Activity Relationships (QSAR) model was developed for predicting the IC50 parameter for factor Xa inhibitors. QSAR have been constructed by combining Genetic Algorithms with Multiple Linear Regressions (GA-MLR) and Support Vector Machine (SVM). The correlation coefficient (R^2) and root mean square error (RMSE) for GA-MLR are 0.895, 0.142 and for SVM are 0.994, 0.032, respectively. The obtained results indicated that the SVM method is superior over the GA-MLR method. Also, molecular docking was used to examine the results more accurately, in which energy interactions were investigated. Molecular docking indicated the binding relationship between the receptor and the ligand. The results showed that reducing the parameters such as electronegativity, and the size of the atoms, as well as increasing the number of loops and groups attached to the structures, is effective in the model.

Keywords Factor Xa · Molecular docking · Quantitative structure-activity relationships · Blood coagulation · Drug design

Introduction

The factor Xa is a serine protease, is synthesized in the liver, and plays an important role in the blood coagulation system [1, 2]. It is a molecular target and it is used to develop anticoagulant drugs for the thromboembolic disorders treatment which leads to morbidity and mortality worldwide [3]. Therefore, factor Xa inhibition is an efficient strategy to treat thromboembolic diseases [4]. The anticoagulant drugs such as warfarin and the heparin series are commonly used to inhibit the factor Xa activities [5]. New drugs design is very expensive and time consuming. Thus, the use of an advanced computational technique is very useful and efficient. Quantitative structure–activity relationship (QSAR) methodology is a statistical method that used to describe the correlation between biological activities and physicochemical properties of compounds (such as size, shape of the molecule, electronegativity, polarizability, van der Waals, atomic mass, etc.) [6]. QSAR studies help us better identify the physicochemical features of inhibitor molecules. This method is a useful and efficient technique for the prediction of biological activities of molecules, especially in drug design [7]. For example, a QSAR model was presented to investigation of molecular features of nucleoside derivatives governing the hTK1 kinase activity [8]. In another work, structure-based drug design method was used for designing new steroid sulfatase inhibitors. [9].

This method is used to find the best candidate of anti-factor Xa drugs, and is carried out on different groups of chemicals, such as 1H-pyrazolo[4,3-d]pyrimidin-7(6H)-one [10], indole/benzoimidazole-5-carboxamidines [11], 2-carboxyindole [12], thiophene derivatives [13], 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid [14], and N2-aroyl anthranilamide [15] inhibitors. QSAR studies help us better understand the structural features of inhibitor molecules relating to inhibitory activities. Different

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computational techniques including comparative molecular field analysis (COMFA) [16], support vector machines (LS-SVM) [17–20], artificial neural networks (ANNs) [21], and comparative molecular similarity analyze (COMSIA) [22] have been used to analyze this correlation. In this research, genetic algorithms multiple linear regression (GA-MLR) is used [23]. One of the methods that is used widely nowadays for categorizing issues is the SVM [24] method which shows good performance in comparison to the ANN method. It is a new regression method which can be implemented for developing a nonlinear model and for the improvement of the GA-MLR model performance. Molecular docking is a computational method that identifies the binding energy of molecules well; in fact, the docking studies are for determining the interconnection between two molecules and finding the best ligand connection to receptor with minimum energy [25].

In the present study, the main aim was to present a new QSAR model consisting of 36 compounds, a novel class of N,N′-disubstituted aroylguanidine-based lactam derivatives to predict IC50 parameters in order to inhibit factor Xa activity [26]. SVM (nonlinear) and MLR (linear) methods are compared with each other. The statistical parameters indicate better results in SVM in comparison to the MLR method. These parameters show that the SVM method has a high ability to predict accurately. All obtained descriptors are interpretable and can verify the validity of the final QSAR models. Also, molecular docking is conducted to investigate the interactions between the ligands and the receptor in the binding site, and to obtain their binding energy [27]. The results of molecular docking and QSAR modeling are complementary and in the same direction.

Materials and methods

Dataset

The dataset for the QSAR modeling study consists of 36 compounds, a novel class of N,N′-disubstituted aroylguanidine-based lactam derivatives, were obtained from Yan Shi's paper [26]. The collected structures of 36 aroylguanidine derivatives which are factor Xa inhibitors are shown with their pIC_{50} values in Table 1. IC_{50} values for aroylguanidine derivatives change from 2 nm up to 110 nm, which are converted to the logarithmic scale pIC_{50} [$-\log IC_{50}$ (nm)] and used for QSAR analyses. For external validation of the models, the dataset is divided in to two series: 7 compounds are selected as the test set and 29 compounds as the training set.

Molecular docking

Molecular docking is a computational technique that predicts the orientations and binding of a ligand to a macromolecule (receptor) when connected to each other in the best binding site for the sustainable complex with the minimum of energy [28].

In this study, the ligands structure is drawn and optimized in the ChemDraw program. To figure out the orientation of the ligands and receptor, the 3D x-ray structure of the compound no. 22 was downloaded from the protein data bank (PDB ID code = 3K9X). Receptor structure preparation was done by means of the AutoDockTools and also by finding the docking site which are used in smina-AutoDock program. For receptor preparation, polar hydrogen's is added and water molecules are deleted, and the Kollman charge [29] is computed for the receptor and saved in pdbqt format. In order to find the binding pocket, the existing amino acids in the binding site are found and then determined on the receptor. A grid box with $28 \times 28 \times 28$ Å dimensions is created which covers all of the determined amino acids. After that, the prepared receptor, ligands, and the grid box dimensions are the AutoDock smina program inputs. The binding affinity for all ligands were calculated.

Geometry optimization and molecular descriptors calculation

In the first step, molecular structures are pre-optimized using the molecular mechanic's force field (MM+) with Hyperchem [30]. Optimized structures were re-optimized using PM6 method with Gaussian software [31]. Then the DRAGON 7 [32] software is used to calculate the molecular descriptors. The DRAGON software calculates 4885 theoretical descriptors for each molecule. These descriptors are categorized to different series such as RDF descriptors, 3D-MoRSE descriptors, and GETAWAY descriptors. Since a large number of these descriptors reveal the ambiguous and incorrect numerical values, 320 equal and near equal variables are removed, also the variables with pair correlation higher than 0.98 (500 variables) are eliminated from the data matrix. Finally, after the applied limitations, 950 theoretical molecular descriptors for each molecule are remained that include 3D-MoRSE, RDF, 3D autocorrelations, and GETAWAY descriptors [33, 34].

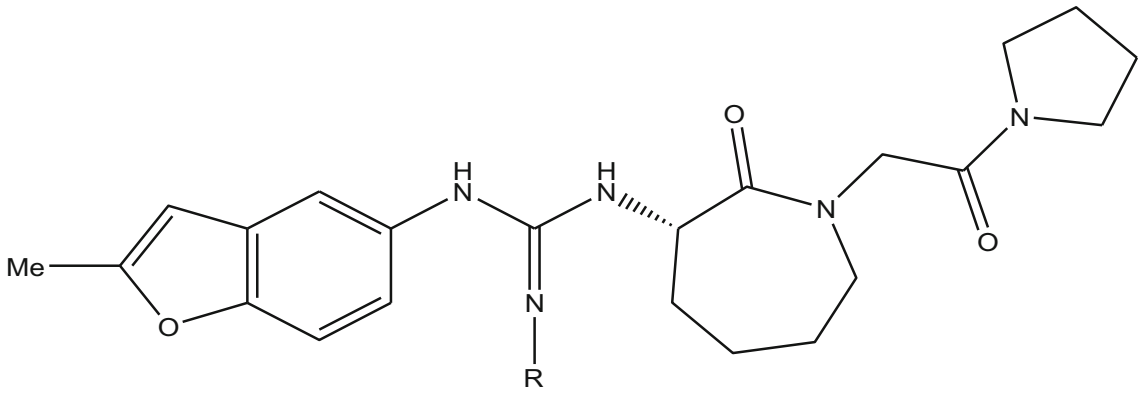
Multiple linear regression and genetic algorithm

In the GA-MLR method, the aim is to find the best correlation between the IC50 values, and a certain number of molecular descriptors which have one or more of the following conditions are removed from the data matrix:

1. Descriptor that have identical values or zero value
2. Its correlation coefficient is more than 98%
3. Its calculation is difficult or descriptor is imprecise.

Then, the remaining descriptors which are collected in a n.m data matrix are used to develop the QSAR model, where

Table 1 Chemical structures and experimental and predicted pIC50 values by GA-MLR and LS-SVM methods



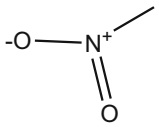
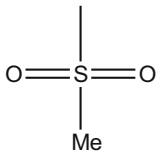
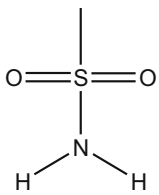
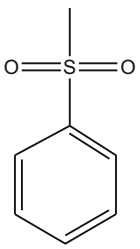
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
3		7.92	7.75	7.91
4		7.60	7.51	7.60
5		8.0	7.92	7.98
6^t		7.92	8.02	7.92
7^t		7.61	7.66	7.61

Table 1 (continued)

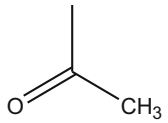
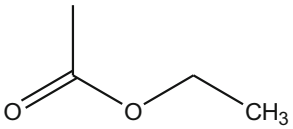
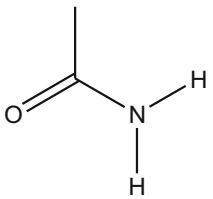
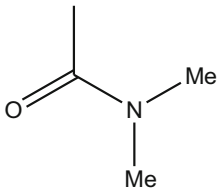
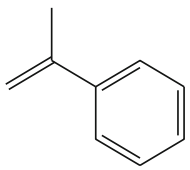
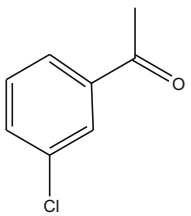
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
8		7.76	7.96	7.76
9^t		7.49	7.52	7.49
10		7.55	7.45	7.54
11		7.82	7.85	7.84
12		8.22	8.15	8.20
13		8.15	8.11	8.14

Table 1 (continued)

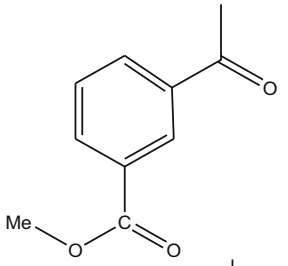
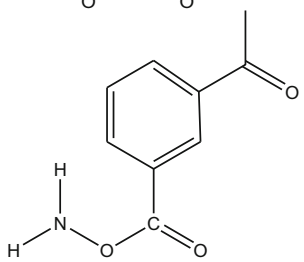
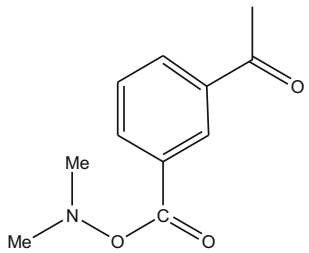
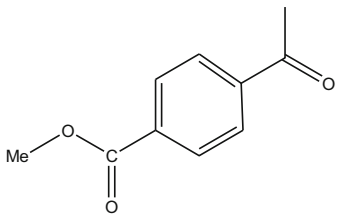
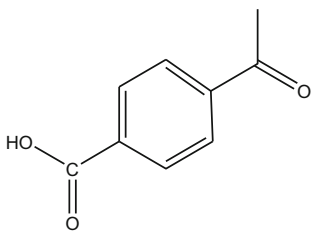
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
14		8.30	8.19	8.30
15[t]		7.74	7.82	7.73
16[t]		8.00	8.00	7.99
17		8.22	8.34	8.24
18		8.69	8.8	8.68

Table 1 (continued)

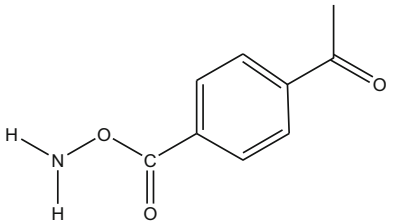
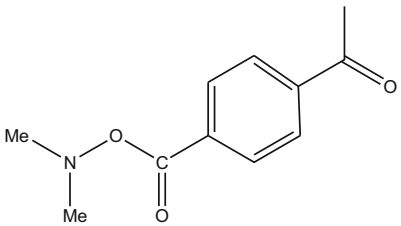
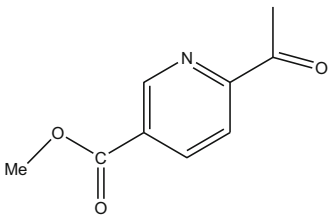
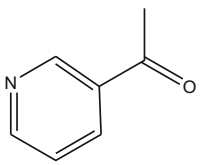
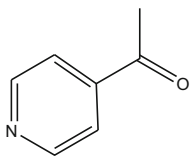
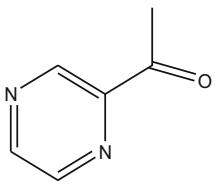
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
19		8.52	8.45	8.50
20		8.52	8.33	8.51
21		7.25	7.40	7.25
22		8.39	8.25	8.38
23		8.52	8.44	8.44
24		7.30	7.26	7.29

Table 1 (continued)

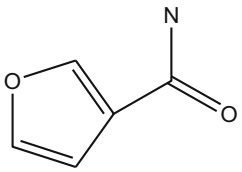
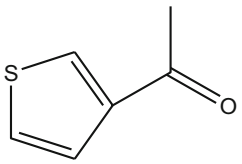
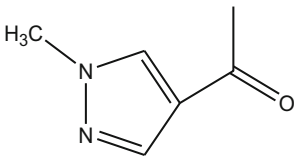
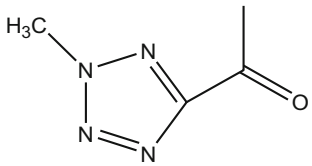
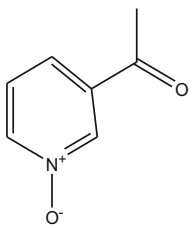
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
25[t]		8.39	8.30	8.35
26[t]		8.22	8.29	8.25
27		8.04	7.87	8.03
28		7.25	7.17	7.24
29		8.00	8.06	7.99

Table 1 (continued)

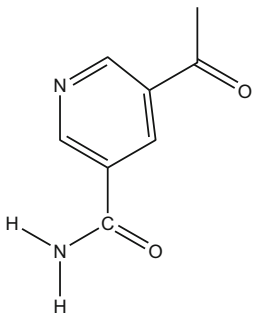
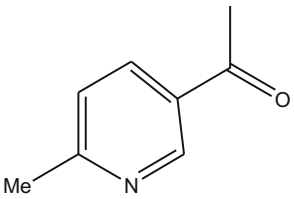
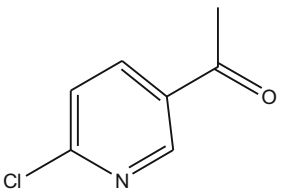
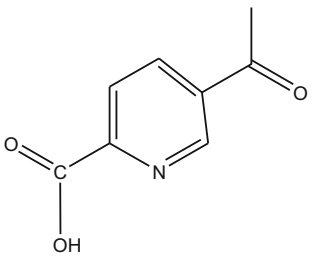
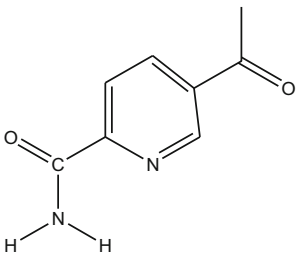
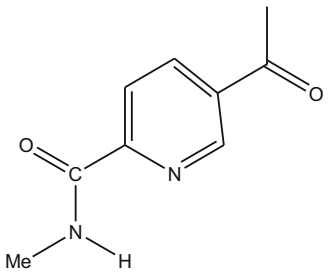
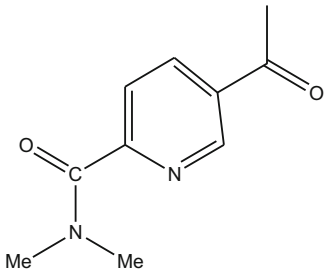
Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
30		7.76	7.94	7.79
31		8.39	8.24	8.33
32		8.30	8.40	8.40
33		8.48	8.64	8.48
34		7.92	8.15	7.93

Table 1 (continued)

Comp.	R	Experimental pIC_{50}	Predicted pIC_{50}	
			GA-MLR	LS-SVM
35		8.22	8.07	8.15
36		8.04	8.11	8.06

t Test set

$n = 36$ and $m = 950$ are respectively the numbers of the compounds and the molecular descriptors [26]. There are different methods for the selection of effective descriptors. Genetic Algorithm is a new method that is used for optimization of problems which their main idea is obtained from the evolution theory of Darwin [26, 34]. GA method is based on the evolution of a population or the primary generation of models in which models are sorted according to an objective function. In GA, each member of the population is described by a chromosome, such that each chromosome gene is occupied by a descriptor. In this algorithm, there are three steps, including Random selection of the first generation, Crossover or Recombination, and Mutation [35]. After obtaining the suitable descriptors by means of the Genetic Algorithm, the selected variables are used to develop the QSAR model.

Least squares-support vector machine method

Least squares-support vector machine (LS-SVM) is a more complete and better version of LS-SVM. LS-SVM is a new regression method which can be implemented for developing a nonlinear model, and improvement of the GA-MLR model performance. The descriptors resulted from both linear and nonlinear methods can be used for developing the QSAR models. For using LS-SVM, a kernel function which obeys some mathematical properties is required. In order to have better results and a smaller number of parameters, Radial Basis Function (RBF) is used as the kernel function. This function has two important variables: γ (regularization factor) and σ^2 (RBF kernel parameter). In this LS-SVM model, the optimum quantities of parameters $\sigma^2 = 4.77$ and $\gamma = 1426$ and other parameters are resulted from MATLAB software calculations [36, 37].

Results and discussion

Molecular modeling

The chemical structures of a novel class of N,N-disubstituted aroylguanidine-based lactam derivatives and their experimental pIC_{50} values are shown in Table 1. Compounds 1 and 2 are drawn separately in Fig. 1 because of having different structures.

Genetic Algorithm was used for the selection of effective descriptors. To obtain the optimum number of descriptors in a model, a graph showing the relationship between the number of descriptors and statistical parameters (R^2 and Q^2), and the effect of this number on the quality of the final model is plotted and shown in Fig. 2. As shown in the figure, R^2 and Q^2 values

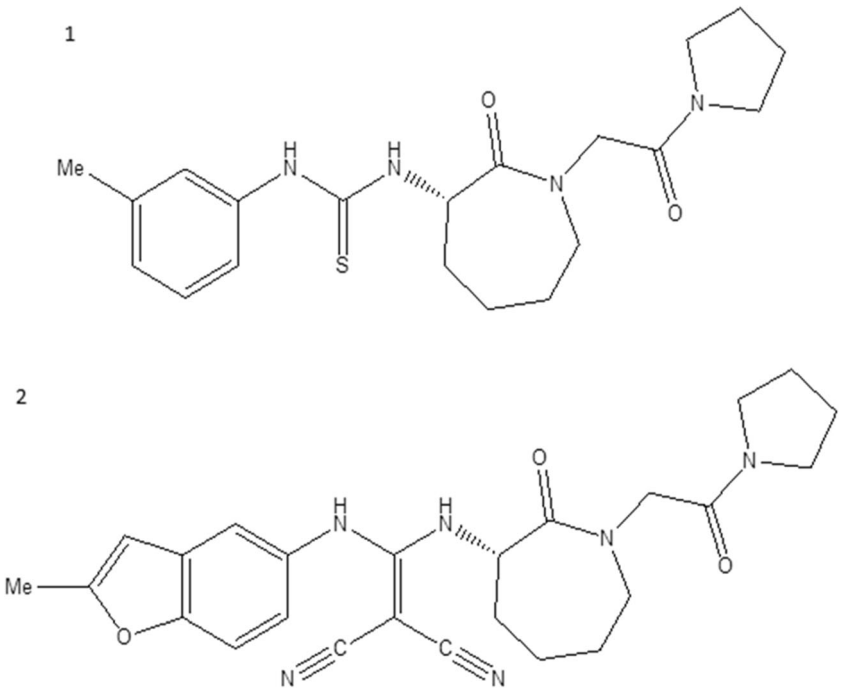


Fig. 1 Chemical structure of compounds 1, 2

increase with a constant trend while the number of descriptors increases from 1 to 5, and then decreases. This shows that the optimum number of descriptors is 5, which is the breaking point, and using more than 5 descriptors leads to complexity of the model [26]. Categorizing the data into training and test sets is an important step to investigate the prediction power of the QSAR modeling. By using the PCA method, it is observed that 81.68% of all variables' information is located in PC1 and PC2, as shown in Fig. 3. The data are distributed in 4 different districts and seven of them are selected as the test set to evaluate the model's prediction power, and the rest of the data ($n = 29$) are used as a training set to generate the model.

Experimental and predicted values of pIC50 by MLR and SVM methods are presented in Table 1. It is shown that predicted pIC_{50} values are near to their experimental values, but the SVM method has lower deviation than MLR in prediction. The experimental pIC_{50} values in comparison with their predicted values are indicated in Fig. 4. These values are

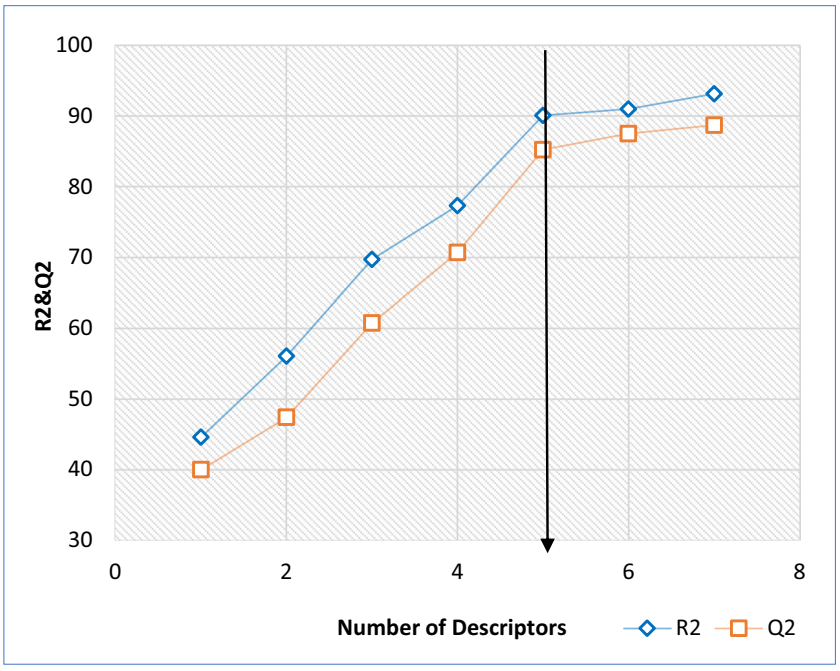


Fig. 2 Plot of statistical parameters (R^2 and Q^2) versus the number of descriptors

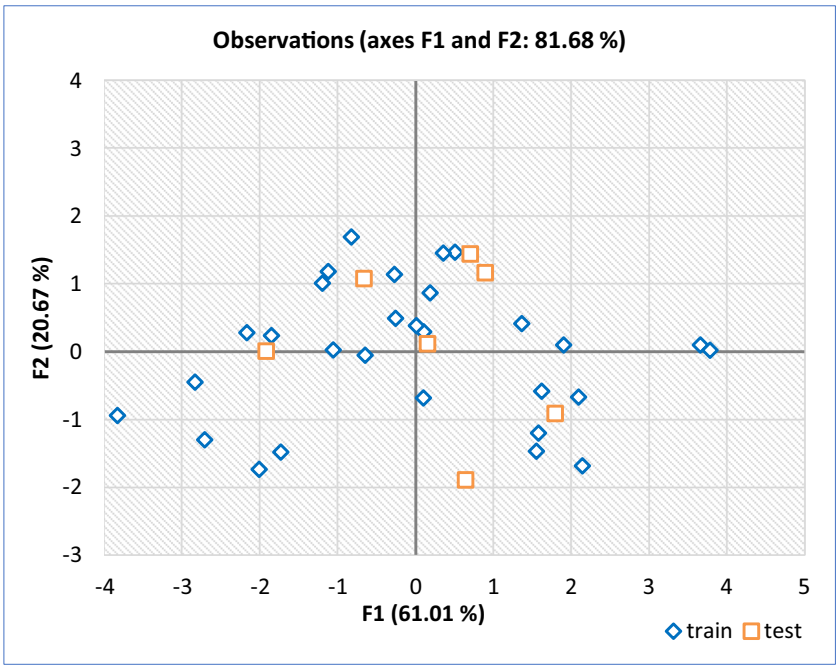


Fig. 3 Principle component analysis for ionic liquids in training and test sets

distributed around a 45° line which indicates the power of the model, and that the SVM model is more efficient than the MLR. Figure 5 shows the diagram of residual values versus experimental values of pIC_{50} for both methods of SVM and MLR. The relatively uniform dispersion of the residuals around the zero line indicates that there is no systematic error in the proposed model.

The obtained model is as the following equation:

$$pIC_{50} = 12.487 - 0.932(TDB05e) - 0.777(RDF110u) + 0.053(Mor20s) - 0.935(H6u) + 0.991(H7s)$$

$$N = 36, R^2 = 0.895, F = 39.495, RMSE = 0.142$$

The different statistical parameters of both MLR and SVM models (including R^2, RMSE, and Q^2) are indicated in Table 2. These parameters are used to investigate the ability of models for estimating and creating of the final model. Also, the Q^2 parameter is achieved to assess the internal validation of the QSAR model. According to the results shown in Table 2, the SVM method parameters are better than those obtained from the MLR method.

Y-scrambling test is used to evaluate the validity and robustness of the QSAR model. In this test, the dependent variable is randomly scrambled between the different molecules and some new QSAR models are built based on previous descriptors [38]. This method is repeated and it is expected that the obtained QSAR model has low R^2 values compared to the original model. Otherwise, the new model is poor and inefficient. In this work, Y-scrambling was run 10 times, and the results are summarized in Table 3. The average value of R^2 is 0.11 that means the predicted models are not accidental.

The applicability domain

The applicability domain (AD) is a theoretical area from chemical space, which is defined by model descriptors, and is indicated using the nature of chemical compounds in the training set and also, the molecular descriptors of each model. Points out of the applicability domain of the model are determined as outliers. The compounds leverage measurement and the William plot [39] are the most important methods investigated in the applicability domain. The leverage amount determines the distance of compounds from the variables space X [40]. The William plot represents the relation between the standardized residuals (R) and the leverage (h) value. This plot investigates whether the new chemical compounds are within the structural range of modeled compounds. Leverage threshold (h^*) is shown via a vertical line which is calculated through the following formula:

$$h^* = 3(p+1)/n$$

Where n is the number of training data and p is the number of descriptors. The William plot is shown in

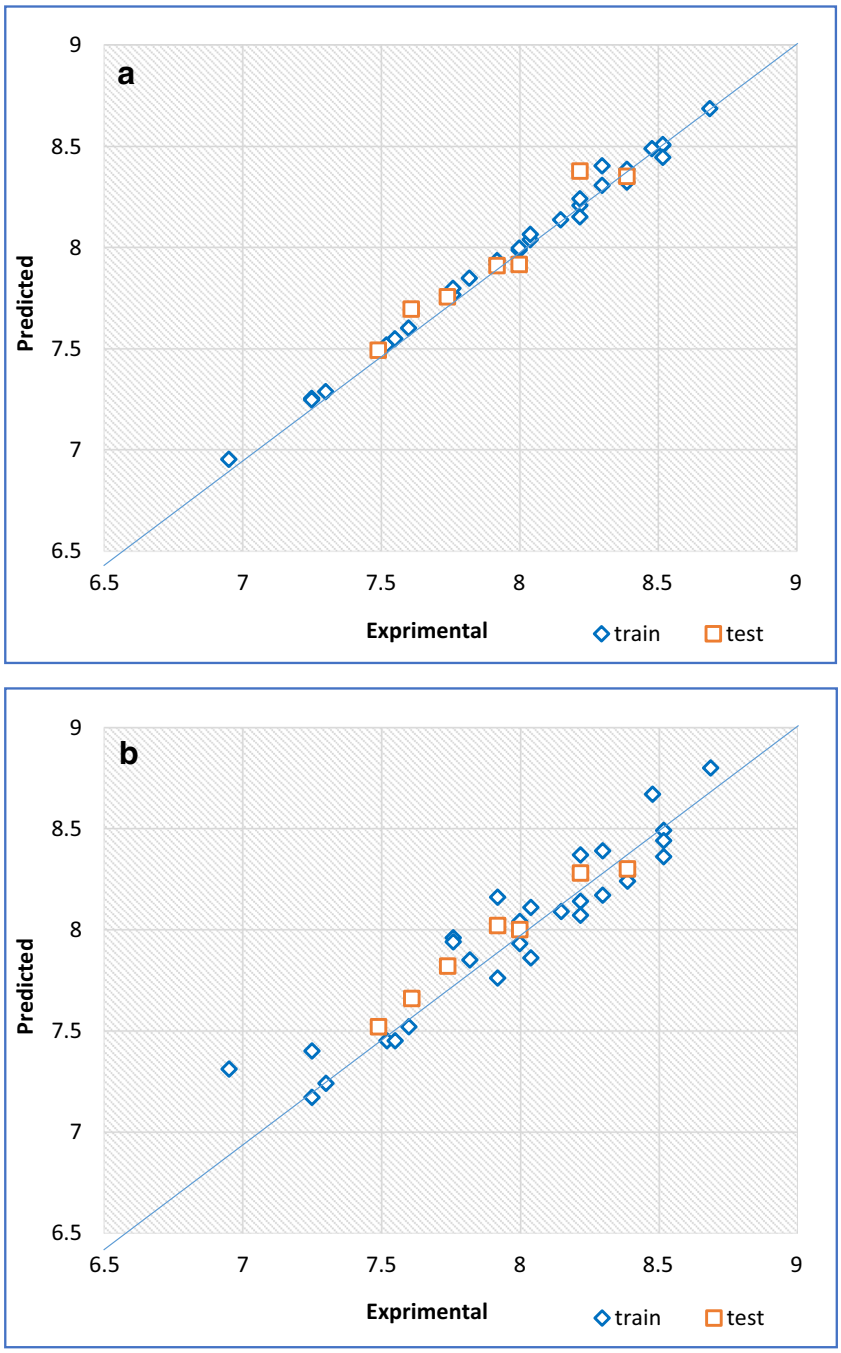


Fig. 4 The experimental values versus predicted values by **a** LS-SVM method and **b** MLR method

Fig. 6 to visualize the AD of MLR and SVM methods. In this dataset, the h^* value is 0.62. The compounds with standardized residual greater than 3 and $h > h^*$ are considered as an outlier compound. According to Fig. 6, there is no compound with a leverage value more than leverage threshold (h^*) in both methods. Compound no. 32 is located on the critical horizontal line in the SVM, and is higher than in the MLR model. This is probably because of the different mechanism of compound 32, or can be attributed to the incorrect experimental input data rather than the molecular structure.

Descriptors interpretation

In Table 4, the correlation matrix among the selected descriptors is shown. It indicates that the correlation coefficient value between these variables is less than 0.6, and this means that the descriptors are independent of each other. Which, according to the results, two descriptors H7s and H6u with correlation coefficient values of 0.698 are more dependent on each other. The selected descriptors are TDB05e, RDF110u, Mor20s, H6u, and H7s. The mean effect shows the relative effect of each descriptor in comparison to other descriptors in

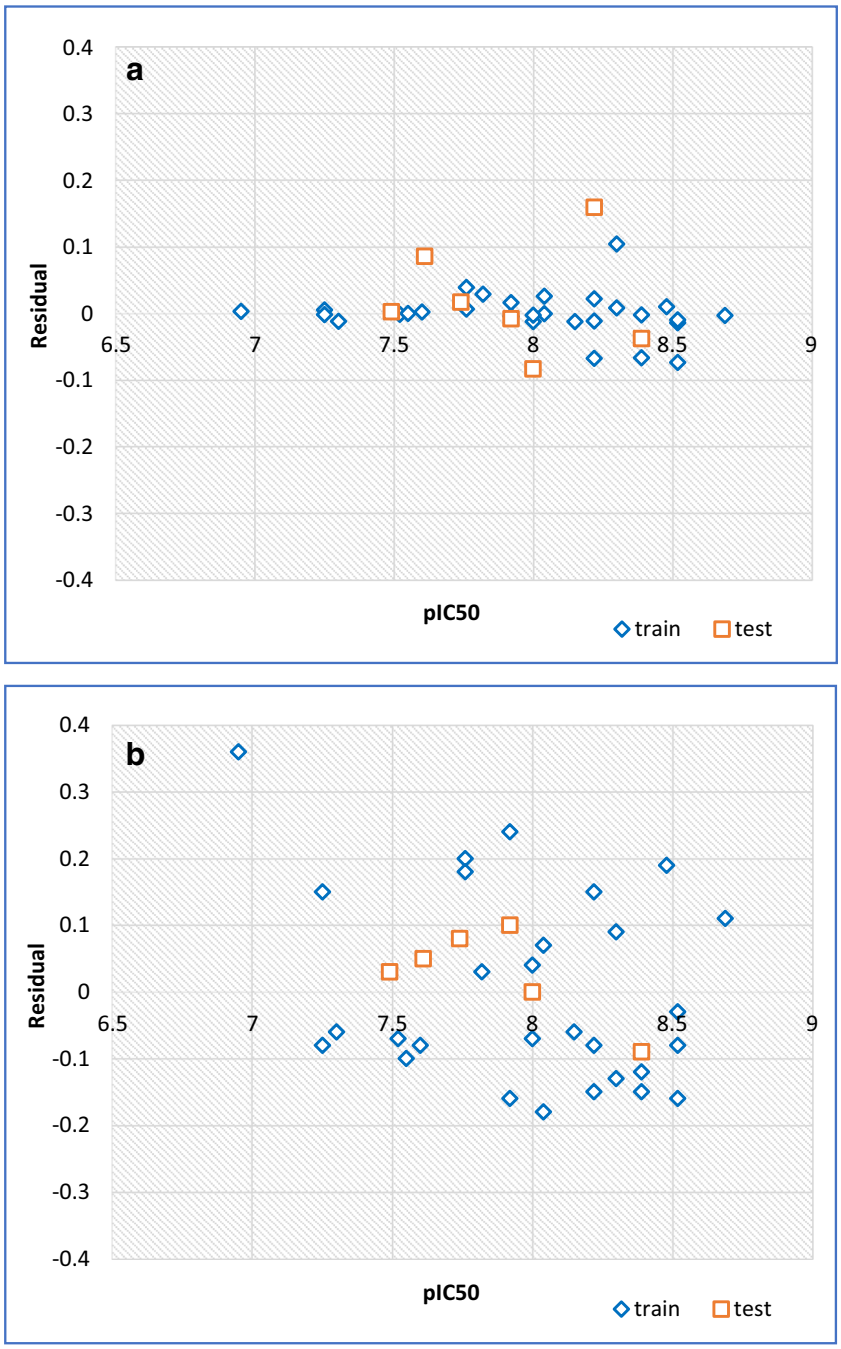


Fig. 5 Plot of residual versus experimental values by **a** LS-SVM method and **b** MLR method

the model. Positive and negative effects (directly and inversely effects) of these variables are indicated in Fig. 7. Although QSAR modeling indicates the importance of statistical parameters, it should present a useful and acceptable concept for the mechanism of inhibitory activity for each descriptor.

Table 2 Statistical parameters for two different models

	Training RMSE	R^2	F	Q^2	Test RMSE	R^2	Overall RMSE	R^2	F	Q^2
MLR	0.142	0.895	39.495	0.825	0.078	0.931	0.131	0.900	54.557	0.852
SVM	0.032	0.994	841.792	0.992	0.077	0.933	0.028	0.995	1255.416	0.994

Table 3 R^2 values for Y-scrambling test

Iteration	R^2 train
1	0.129
2	0.094
3	0.120
4	0.088
5	0.251
6	0.087
7	0.009
8	0.095
9	0.125
10	0.104

The first and second significant descriptors are the GETAWAY; the GETAWAY descriptors are geometrical descriptors intending to indicate the adaptation of molecule 3D geometry, atoms correlations, and their chemical information [41]. H7s (H autocorrelation of lag 7 / weighted by l-state) is the most effective descriptor, and provides information on the structural fragments, such as the effective position of substituents and fragments in the molecular space and information of molecule size and shape [42]. This descriptor has a positive sign and the greatest effect on the final model. H6u is the second significant descriptor. H6u

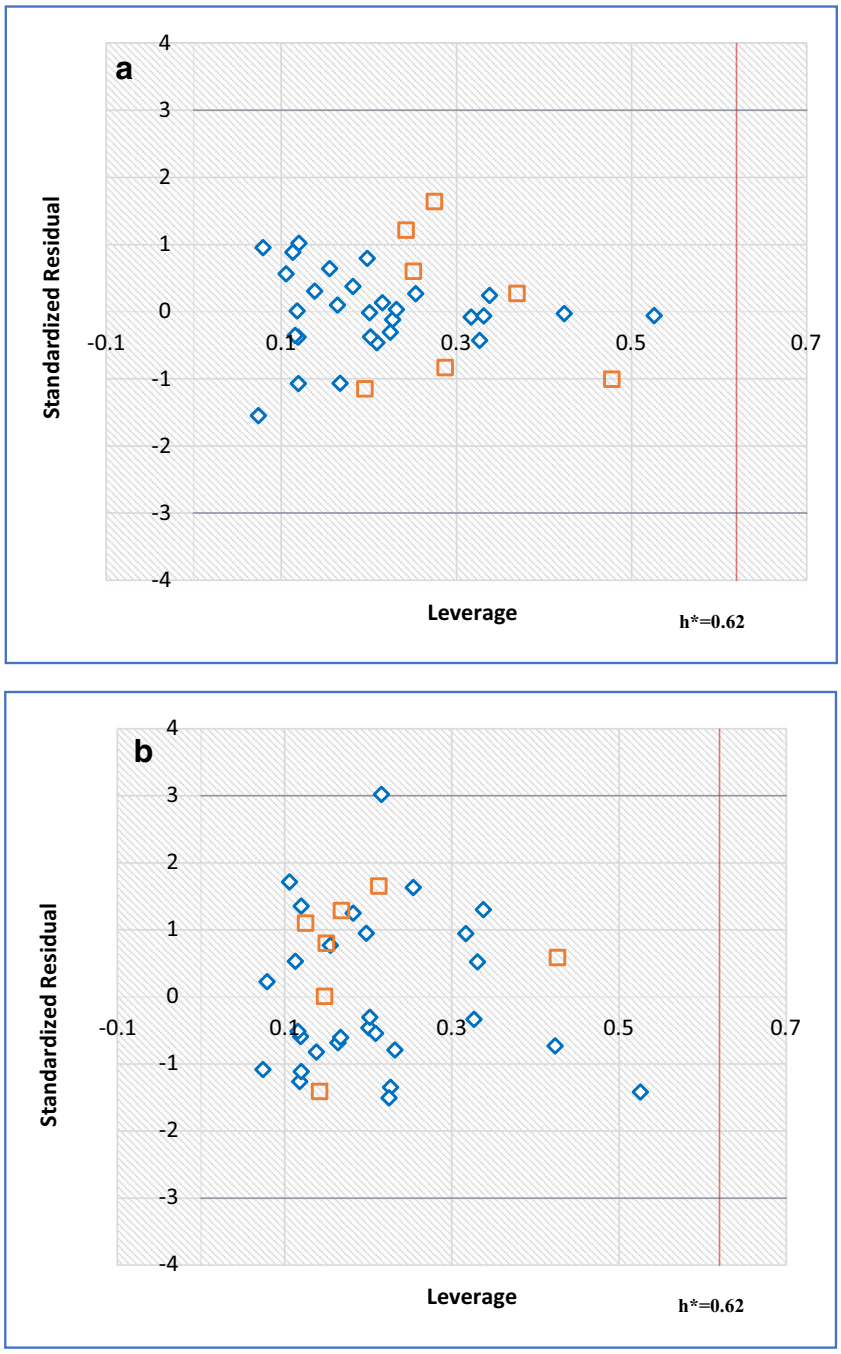


Fig. 6 Williams plot by **a** LS-SVM method and **b** MLR method

Table 4 Correlation coefficient between selected molecular descriptors

Variables	TDB05e	RDF110u	Mor20s	H6u	H7s
TDB05e	1	− 0.507	− 0.627	− 0.629	− 0.371
RDF110u	− 0.507	1	0.494	0.551	0.408
Mor20s	− 0.627	0.494	1	0.266	0.154
H6u	− 0.629	0.551	0.266	1	0.698
H7s	− 0.371	0.408	0.154	0.698	1

(H autocorrelation of lag 6 / unweighted) is related to the size and location of the atom in the molecule. Since this descriptor has a negative sign, it has an inverse relation with pIC50. Therefore, via increasing the size of the atom and the distance between an atom and the center of the molecule, the value of this descriptor increases [43]. The next significant descriptor is TDB05e (3D Topological distance-based descriptors − lag 5 weighted by Sanderson electronegativity) [44]. This is one of the three-dimensional geometric descriptors associated with spatial correlations between the atoms or points on the surface of molecules. TDB05e has a negative sign which indicates that pIC50 is inversely related to this descriptor. The fourth descriptor is RDF110u (Radial Distribution Function − 110 / unweighted), a kind of radial distribution function based on the distance distribution in the molecule, and its concept is the probability of finding an atom in a spherical volume with radius R [45]. Also, the RDF110u is inversely related to this descriptor. The last selected descriptor is Mor20s (Signal 20 / weighted by l-state). This descriptor is one of the 3D-descriptors related to electron diffraction which explains the 3D coordination of the different atoms in a molecule [46]. Mor20s has a positive sign indicating the direct relation of pIC50 with this descriptor, and also, that it has the least effect on the final model.

To summarize, by decreasing parameters such as electronegativity, size of the atom, the distance between the atom and the center of the molecule, and value of radial distribution function, and also by increasing the 3D coordination of the different atoms in a molecule and the rings and groups attached to the structure of the molecules and their deformation, a better inhibitory activity level of pIC_{50} (high value) is obtained.

Molecular docking studies

To find the ligands binding modes and to determine the amino acids which are located in the ligand and receptor binding site, molecular docking is performed in a ligand-binding pocket. After running the docking program, the binding affinity is obtained for the whole ligand-receptor binding, which are reported in Table 5. The achieved energies are compared with the experimental pIC_{50}. According to the obtained results, compounds 1 and 18 have the lowest and highest value of binding energy, respectively. Figure 8 indicates that ligand no. 18 has established a hydrogen bond with GLN192B amino acids in the active site. Also, the existence of hydrophobic interactions between the ligand and amino acids in the active site leads to an increase in the binding energy of this compound. This indicates that ligand no. 18 has a good orientation toward this molecule. On the other hand, it has the highest amount of pIC_{50}. As a result, the greater the binding energy, the greater the power of inhibition and vice versa.

Recent studies have presented advances in FXa inhibitors through the QSAR method. For example, Alan R. Katritzky et al. [47] developed a QSAR method involving a multi-linear MLR and nonlinear ANN based on 60 benzoxazinone

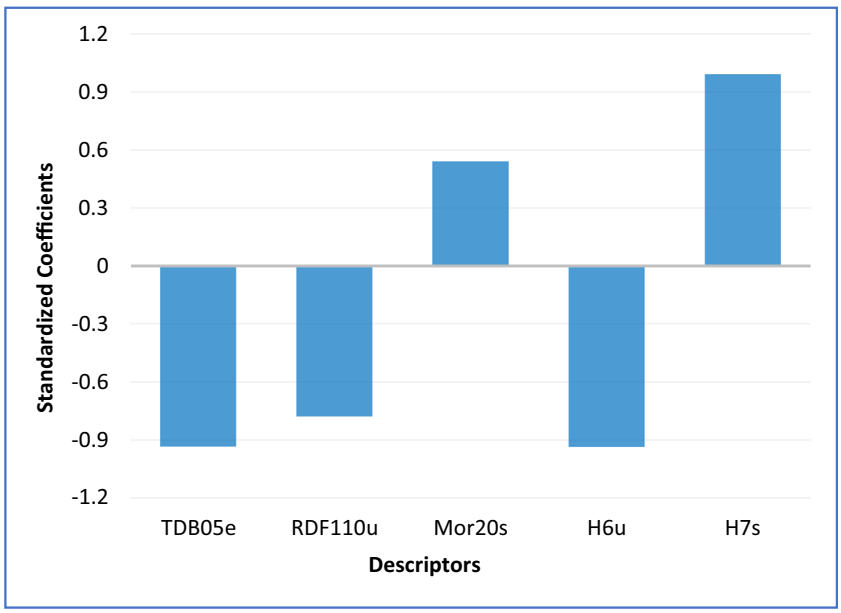


Fig. 7 Relative effect of each descriptor in the MLR model

Table 5 The obtained binding energy by AutoDock

No.	pIC50	Minimized affinity
1	6.95	7.45
2	7.52	7.99
3	7.92	8.37
4	7.6	8.06
5	8	8.94
6	7.92	8.39
7	7.61	8.58
8	7.76	8.05
9	7.49	8.22
10	7.55	8.31
11	7.82	8.82
12	8.22	9.05
13	8.15	9.13
14	8.3	8.96
15	7.74	9.26
16	8	8.44
17	8.22	9.52
18	8.69	9.80
19	8.52	8.39
20	8.52	8.08
21	7.25	7.88
22	8.39	9.06
23	8.52	8.64
24	7.3	8.82
25	8.39	8.51
26	8.22	8.81
27	8.04	8.99
28	7.25	8.68
29	8	8.37
30	7.76	9.32
31	8.39	9.13
32	8.3	9.19
33	8.48	9.22
34	7.92	9.53
35	8.22	8.70
36	8.04	9.33

derivative antiplatelet agents. The obtained descriptors indicate the importance of the C–O bond reactivity in determining the antiplatelet activity of compounds. Some descriptors in this model express the importance of the N atom and CO group for the interaction between the inhibitor and the biological receptor. Comparing the MLR and ANN methods indicates that ANN has a higher prediction power with respect to the R^2 value and the proximity of the experimental and predicted values. The statistical parameters are shown in the Table 6. In another work, Cheng Xu et al. [48] used a 3D-QSAR method to study 38 new tricyclic fused oxazolidinones inhibitors. They used COMFA, COMSIA, and molecular docking methods. According to the statistical results of the two methods COMFA and COMSIA which are presented in the Table 6, COMSIA is more efficient and better than COMFA. In the docking results, four hydrogen bonds were explored in the active site. Also, the steric, hydrophobic, H-acceptor, and electrostatic fields had important roles in the modeling.

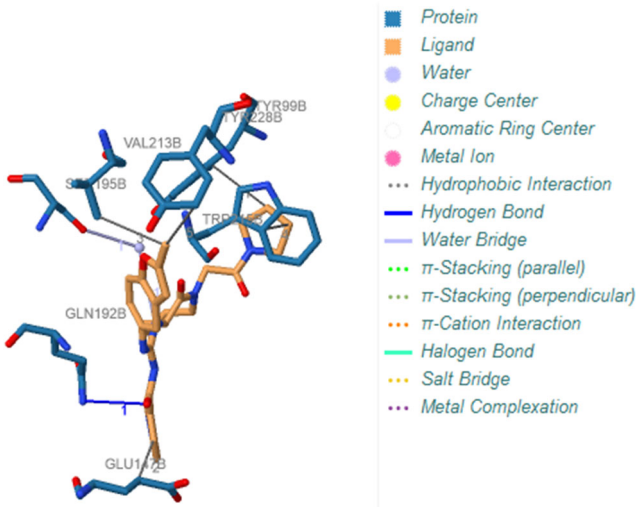


Fig. 8 Interaction between ligand 18 and amino acids in the active site

Table 6 Comparison of the different models for prediction of IC50 for factor Xa inhibitors

	R^2	Q^2	F	S	N_{des}	$N_{dataset}$	Methods
Ref. [45]	0.821	–	49.551	0.621	5	60	MLR
	0.909	–	–	–	5	60	ANN
Ref. [46]	0.984	0.511	240.836	0.174	–	38	COMFA
	0.993	0.700	348.096	0.119	–	38	COMSIA
This work	0.895	0.825	39.495	0.142	5	36	MLR
	0.994	0.992	841.792	0.032	5	36	SVM

According to the obtained results in this research, the SVM method has more prediction power in respect to the other researches in QSAR modeling. As mentioned in Ref. [45], COMSIA is a good method of prediction, but its lower Q^2 leads to a lower prediction power in comparison to SVM. Reviewing the literature shows that the use of the SVM method in QSAR modeling is more appropriate to predict the values of pIC_{50} for factor Xa enzyme inhibitors.

Conclusion

In this research, linear (MLR), nonlinear (SVM), and molecular docking methods are used to construct a QSAR model to predict the inhibitory activities of 36 novel compounds of aroylguanidine derivatives as FXa inhibitors. The best model is obtained with five descriptors that are as follows: TDB05e, RDF110u, Mor20s, H6u, and H7s. For TDB05e and H7s, it can be concluded that electronegativity, size, shape, and specific atomic properties have a great influence on the molecule. On the other hand, Mor20s descriptor shows that the 3D coordination of different atoms in the molecule has the lowest effect on pIC_{50}. The resulted statistical characteristics for MLR and SVM are as follows: $R^2 = 0.895$, $Q^2 = 0.825$, RMSE = 0.142 and $R^2 = 0.994$, $Q^2 = 0.992$, RMSE = 0.032,

respectively. Also, the applicability domain for new chemicals was confirmed by the leverage. Comparison of the results shows that the SVM method is a powerful and reliable method, and it also is more efficient than the method MLR for model prediction. In addition, the results obtained from molecular docking shows that because of hydrogen binding with GLN192B amino acids and also hydrophobic interactions, the ligand no. 18 (with highest pIC50) has the highest binding energy. It can therefore be concluded that this study aids us to better understand the relationship between the structural characteristics of inhibitor molecules and their inhibitory activities. These results can lead to prediction and design of new and potent FXa inhibitors in future.

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Compliance with ethical standards

Conflict of interest The authors declare that there are no conflicts of interest regarding the publication of this paper.

References

1. Eisenberg PR, Siegel JE, Abendschein DR, Miletich JP (1993) Importance of factor Xa in determining the procoagulant activity of whole-blood clots. J Clin Investig 91:1877–1883
2. Shi Y, Sitkoff D, Zhang J, Han W, Hu Z, Stein PD, Wang Y, Kennedy LJ, Oconner SP, Ahmad S, Liu EC, Seiler SM, Lam PY, Robl JA, Macor JE, Atwal KS, Zahler R (2007) Amino (methyl) pyrrolidines as novel scaffolds for factor Xa inhibitors. Bioorg Med Chem 17:5952–5958
3. Xing J, Yang L, Li H, Li Q, Zhao L, Wang X, Zhang Y, Zhou M, Zhou H, Zhang H (2015) Identification of anthranilamide derivatives as potential factor Xa inhibitors: drug design, synthesis and biological evaluation. Eur J Med Chem 95:388–399
4. Iwatsuki Y, Sato T, Moritani Y, Shigenaga T, Suzuki M, Kawasaki T, Funatsu T, Kaku S (2011) Biochemical and pharmacological profile of darexaban, an oral direct factor Xa inhibitor. Eur J Pharmacol 673:49–55
5. Verma RP, Hansch C (2010) QSAR modeling of taxane analogues against colon cancer. Eur J Med Chem 45:1470–1477
6. Choudhari PB, Bhatia MS, Kumbhar SS (2012) 3D QSAR and pharmacophore modeling of selected factor Xa inhibitors. Med Chem Res 21:1427–1432
7. Jain SV, Ghate M, Bhadoriya KS, Bari SB, Chaudhari A, Borse JS (2012) 2D, 3D-QSAR and docking studies of 1,2,3-thiadiazole thioacetanilides analogues as potent HIV-1 non-nucleoside reverse transcriptase inhibitors. Org Med Chem Lett 2:2–22
8. Jagiello K, Makurat S, Perec S, Rak J, Puzyn T (2018) Molecular features of thymidine analogues governing the activity of human thymidine kinase. Struct Chem 29:1367–1374
9. Jagiello K, Sosnowska A, Kar S, Demkowicz S, Dasko M, Leszczynski J, Rachon J, Puzyn T (2017) Geometry optimization of steroid sulfatase inhibitors—the influence on the free binding energy with STS. Struct Chem 28:1017–1032
10. Fevig JM, Cacciola J, Buriak J, Rossi KA, Knabb RM, Luettgen JM, Wong PC, Bai SA, Wexler RR, Lam PY (2006) Preparation of 1-(4-methoxyphenyl)-1H-pyrazolo [4, 3-d] pyrimidin-7(6H)-ones as potent, selective and bioavailable inhibitors of coagulation factor Xa. Bioorg Med Chem Lett 16:3755–3760
11. Bhongade BA, Gouripurb VV, Gadada AK (2005) 3D-QSAR CoMFA studies on trypsin-like serine protease inhibitors: a comparative electivity analysis. Bioorg Med Chem 13:2773–2782
12. Nazare M, Will DW, Matter H (2005) Probing the subpockets of factor Xa reveals two binding modes for inhibitors based on a 2-carboxyindole scaffold: a study combining structure-activity relationship and X-ray crystallography. J Med Chem 48:4511–4525
13. Dguigui K, Elhallaoui M (2013) 3D-QSAR modeling of substituted thiophene-anthranilamides as potent inhibitors of human factor Xa using quantum chemical descriptors. Int J Sci Res 4:1237–1247
14. Al-Horani RA, Mehta AY, Desai UR (2012) Potent direct inhibitors of factor Xa based on the tetrahydroisoquinoline scaffold. Eur J Med Chem 54:771e783
15. Mendel D, Marquart AL, Joseph S (2007) Anthranilamide inhibitors of factor Xa. Bioorg Med Chem Lett 17:4832–4836
16. Cramer RD, Patterson DE, Bunce JD (1988) Comparative molecular field analysis (CoMFA). 1. Effect of shape on binding of steroids to carrier proteins. J Am Chem Soc 110:5959–5967
17. Vapnik V (1995) The nature of statistical learning theory. Springer-Verlag, New York
18. Rezaei B, Riahi S, Ebrahimpoor Gorgi A (2020) Molecular investigation of amine performance in the carbon capture process: least squares support vector machine approach. Korean J Chem Eng 37: 72–79
19. Khaheshi Hasnkiadeh SH, Riahi S, Mohammadi-khanaposhti M, Shokrollahzadeh H (2019) Prediction of amines capacity for carbon dioxide absorption based on structural characteristics. Ind Eng Chem Res 58:8763–8771
20. Mehraein I, Riahi S (2017) The QSPR models to predict the solubility of CO_2 in ionic liquids based on least-squares support vector machines and genetic algorithm-multi linear regression. J Mol Liq 225:521–530
21. Claycamp HG, Sussman NB, Macina O, Rosenkranz HS (1999) Artificial neural networks as statistical tools in SAR/QSAR modeling. AAAI Technical Report SS-99-01, 1-4
22. Klebe G, Abraham U, Mietzner T (1994) Molecular similarity indices in a comparative analysis (CoMSIA) of drug molecules to correlate and predict their biological activity. J Med Chem 37: 4130–4146
23. Kovalishyn V, Tanchuk V, Charochkina L, Semenuta I, Prokopenko V (2012) Predictive QSAR modeling of phosphodiesterase 4 inhibitors. J Mol Graph Model 32:32–38
24. Tian Y, Shi Y, Liu X (2012) Recent advances on support vector machines research. Technol Econ Dev Econ 18:5–33
25. Meng XY, Zhang HX, Mezei M, Cui M (2012) Molecular docking: a powerful approach for structure-based drug discovery. Curr Comput Aided Drug Des 7:146–157
26. Shi Y, Li C, Zhang J, Shi M, Huang C, Sitkoff D (2009) Aroylguanidine-based factor Xa inhibitors: the discovery of BMS-344577. Bioorg Med Chem 19:6882–6889
27. Sepehri B, Rasouli Z, Ghavami R (2016) Molecular docking and QSAR analysis of naphthyridone derivatives as ATAD2 bromodomain inhibitors: application of CoMFA, LS-SVM, and RBF neural network. Med Chem Res 25:2895–2905
28. Fatemi MH, Heidari A, Gharaghani S (2015) QSAR prediction of HIV-1 protease inhibitory activities using docking derived molecular descriptors. J Theor Biol 369:13–22
29. Morris GM, Huey R, Lindstrom W, Sanner MF, Belew RK, Goodsell DS, Olson AJ (2009) AutoDock4 and AutoDockTools4: automated docking with selective receptor flexibility. J Comput Chem 30:2785–2791
30. Release H (2002) 7.5 for Windows, molecular modeling system, Hypercube. Inc. http://www.Hyper.Com

31. Frisch MJ, Trucks GW, Schlegel HB, Scuseria GE, Robb MA, Cheeseman JR, Montgomery JJA, Vreven T, Kudin KN, Burant JC, Millam JM, Iyengar SS, Tomasi J, Barone V, Mennucci B, Cossi M, Scalmani G, Rega N, Petersson GA, Nakatsuji H, Hada M (2009) Gaussian 09. Gaussian, Inc., Wallingford
32. Todeschini R, Consonni V, Mauri A, Pavan M (2002) DRAGON-Software for the calculation of molecular descriptors version 2.1
33. Todeschini R, Consonni V (2000) Handbook of molecular descriptors. Wiley VCH, Weinheim
34. Modarresi H, Modarress H, Dearden JS (2007) QSPR model of Henry's law constant for a diverse set of organic chemicals based on genetic algorithm-radial basis function network approach. Chemosphere 66:2067–2076
35. Goldberg DE (1989) Genetic algorithms in search, optimization and machine learning. Addison-Wesley Longman, Boston
36. Yao XJ, Doucet JP, Zhang RS, Chen HF, Liu MC, Hu ZD, Fan BT (2004) Comparative study of QSAR/QSPR correlations using support vector machines, radial basis function neural networks and multiple linear regression. J Chem Inf Model 44:1257–1266
37. Ghaslani D, Es'haghi Gorji Z, Ebrahimpour Gorji A, Riahi S (2017) Descriptive and predictive models for Henry's law constant of CO2 liquids: a QSPR study. Chem Eng Res Des 120:15–25
38. Tropsha A, Gramatica P, Gombar VK (2003) The importance of being earnest: validation is the absolute essential for successful application and interpretation of QSPR models. QSAR Comb Sci 22:69–77
39. Gramatica P (2007) Principles of QSAR models validation: internal and external. QSAR Comb Sci 26:694–701
40. Netzeva TI, Worth AP, Aldenberg T, Benigni R, Cronin MT, Gramatica P, Jaworska JS, Kahn S (2005) Current status of methods for defining the applicability domain of (quantitative) structure–activity relationships. The report and recommendations of ECVAM Workshop 52. Altern Lab Anim 33:155–173
41. Saiz-Urra L, Gonzalez MP, Fall Y, Gomez G (2007) Quantitative structure-activity relationship studies of HIV-1 integrase inhibition. 1. GETAWAY descriptors. Eur J Med Chem 42:64–10
42. Xinliang Y, Xianwei H (2017) A quantitative relationship between Tgs and chain segment structures of polystryrenes. Plimeros, 27: no. 1 Sao Carlos
43. Fatemi M, Gharaghani S (2007) A novel QSAR model for prediction of apoptosis-inducing activity of 4-aryl-4-H-chromenes based on support vector machine. Bioorg Med Chem 15:7746–7754
44. Klein CT, Kaiser D, Ecker G (2004) Topological distance based 3D descriptors for use in QSAR and diversity analysis. J Chem Inf Comput Sci 44:200–209
45. Hemmer MC, Steinhauer V, Gasteiger J (1999) Deriving the 3D structure of organic molecules from their infrared spectra. Vib Spectrosc 19:151–164
46. Schuur JH, Selzer P, Gasteiger J (1996) The coding of the three-dimensional structure of molecules by molecular transforms and its application to structure-spectra correlations and studies of biological activity. J Chem Inf Comput Sci 36:334–344
47. Katritzky AR, Pacureanu LM, Slavov S, Dobchev DA, Karelson M (2006) QSAR study of antiplatelet agents. Bioorg Med Chem 14: 7490–7500
48. Xu C, Ren Y (2015) Molecular modeling studies of [6,6,5] tricyclic fused oxazolidinones as FXa inhibitors using 3D- QSAR, topomer CoMFA, molecular docking and molecular dynamics simulations. Bioorg Med Chem Lett 25:4522–4528

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